	Existing pretrained models	Additional training or modules	Training from scratch
	Compatible	Partially compatible	New model required
RG Surface form to value	• Numeracy / perturbation SFT • R2L separators	• Number Token Loss • Value-aware prefix / ANI • Place-value annotation	• Digit-level tokenization • xVal • FoNE / BitTokens
PG Arithmetic procedures	• Arithmetic / mathematical SFT • Process supervision • Head-targeted tuning	• Reversed output / LEFT • Integrated calculator	• Abacus embeddings • Position coupling • Randomized PE / NoPE • Looped transformers
Inference-time support	CoT · self-consistency · tools (PAL / PoT)		